

Internship Notification Form, IIT Delhi

About Organisation

Name of Company:	Indian School of Business
Date of Establishment:	2001-01-11
Number of Employees:	1000
Social Media Page Link:	https://www.facebook.com/ISBedu/
Website:	https://isb.edu
Type of Organization:	Private
Nature of Business:	Teaching/Research

Internship Profile

Job Title:	Summer Intern
Job Description:	Early detection of climate impacts on crop production
	Number of positions: 2

More than 60% of farmland in India is rainfed, resulting in high volatility in yields in line with variability in monsoon rainfall. Even more critically, rainfed farming dominates the poorest regions of the country, severely affecting the well-being of some of the most vulnerable communities. Experience and anecdotal evidence suggest that some of the most debilitating impacts can be traced back to deviations from the normal in very early stages of the crop cycle – before, during, and up to four weeks after sowing. Presumably, adequate rainfall during germination and vegetative growth enables crops to better withstand impacts such as long dry spells when they occur later in the crop cycle. This project will use a plot-level data on major crops in India to estimate the impact of early-season rainfall deficits and develop a model to predict crop losses using rainfall data from the first 30 days of the crop season. We will build a model for predicting crop yields and use climate variables to estimate losses in bad years relative to years with normal monsoon rainfall. The dataset represents >300,000 observations from ~45,000 farmers across India over 2008-2020, with information on village location, labour and machine costs, costs of inputs such as seeds, fertilizers, and pesticides, as well as irrigation.

Information Infrastructure on Zero Access and Disclosure (IIZAD): Developing a Blockchain-Based Privacy-Preserving Architecture

Number of positions: 2

As the digital landscape continues to evolve, the importance of privacy in decentralized systems has become paramount. This research focuses on the development of a blockchain-based privacy-preserving architecture to address the growing concerns surrounding the security of sensitive data. The aim is to leverage the inherent characteristics of blockchain technology to create a robust and transparent framework that ensures user privacy while maintaining the decentralized nature of the system. The proposed IIZAD architecture employs cryptographic techniques, consensus mechanisms, and smart contract functionalities to establish a secure and privacy-centric ecosystem. By integrating advanced cryptographic primitives such as zero-knowledge proofs, homomorphic encryption, and ring signatures, the system aims to enable confidential transactions, data sharing, and identity protection without compromising the decentralization and transparency aspects of blockchain networks. The research will involve a combination of theoretical analysis, algorithm development, and practical implementation to validate the effectiveness of the proposed privacy-preserving architecture. The outcomes of this research aim to contribute to the advancement of secure and private decentralized systems, providing a foundation for the development of applications that prioritize user privacy in the digital era.

Game For Good (gfG): A Gamified Approach to Analyzing Institutional and Social Interactions

Number of positions: 3

This project investigates the potential of game design as a novel and engaging method for understanding social and institutional behavior. By leveraging the immersive nature of games, this study aims to create a gamified platform that simulates real-world social interactions within institutional settings. The primary focus is on analyzing decision-making processes, cooperation dynamics, and the emergence of social norms within these environments. Key objectives of the research include designing and implementing a gamified simulation environment that replicates key aspects of social institutions, incorporating elements such as roles, responsibilities, and decision-making scenarios. This platform will serve as a controlled space for observing and analyzing human behavior. Additionally, the research involves employing the gamified environment to collect rich behavioral data on how individuals interact within institutional frameworks, studying decision patterns, communication strategies, and the impact of varying institutional structures on participant behavior. Machine learning algorithms will be applied to analyze the vast datasets generated by the gamified simulations, aiming to identify patterns, correlations, and emergent behaviors within the social institutional context. The research

incorporates principles of human-centered design to ensure the gamified platform is intuitive, engaging, and reflective of real-world institutional scenarios. User feedback and iterative design processes will be employed to enhance the authenticity and effectiveness of the simulation. Furthermore, the findings from the gamified simulations will be extended to real-world social institutions, such as workplaces, educational institutions, and governance structures, aiming to draw parallels between simulated and actual behavior. The research outcomes have the potential to contribute to a deeper understanding of social institutional behavior by combining the immersive nature of games with rigorous data analysis techniques, offering a unique perspective on human decision-making within complex social systems and implications for organizational psychology, sociology, and the design of more effective and responsive institutional structures.

Enhancing Biodiversity Monitoring through Immersive Technology

Number of positions: 3

This project introduces an innovative approach to forest species inventory through the development of an augmented reality (AR) platform. The objective is to leverage AR technology to enhance the efficiency and accuracy of field-based biodiversity monitoring, enabling researchers, foresters, and conservationists to conduct real-time species identification and inventory assessments in forest ecosystems. The AR platform integrates machine learning algorithms, computer vision techniques, and species recognition models to provide an immersive and interactive experience for users. Key objectives of the research include designing an intuitive AR interface that overlays real-time information on tree species, including taxonomy, ecological significance, and growth patterns, onto the user's field of view, enhancing the ability to identify and catalog species during field surveys. Additionally, the research involves implementing machine learning algorithms for species recognition using visual data captured through AR devices, training the model on a diverse dataset of plant species to enable accurate and rapid identification in various environmental conditions. Mechanisms are developed to integrate real-time data, such as GPS coordinates, environmental conditions, and timestamped observations, into the AR platform, facilitating the creation of dynamic and geospatially accurate forest species inventories. Collaborative features are incorporated into the AR platform, allowing multiple users to contribute to the inventory in real-time and enabling seamless data sharing, fostering collaborative research efforts and enhancing the collective understanding of forest ecosystems. Extensive field testing is conducted to validate the accuracy and reliability of the AR-based forest species inventory platform, assessing its performance across different forest types, geographical regions, and environmental conditions. The outcomes of this research aim to revolutionize traditional forest species inventory methods

revolutionize traditional forest species inventory methods by providing a user-friendly, technologically advanced solution that has the potential to expedite biodiversity assessments, support conservation efforts, and contribute valuable data for ecological research, while the platform's immersive nature enhances user engagement and fosters a deeper connection between researchers and the natural environment.

Consensus Conundrums: Cellular Automata Games for Navigating the Consent Maze

Number of positions: 2

This project aims to innovate in the realm of community consent for information sharing by integrating game design principles with cellular automata. The objective is to create an engaging and participatory game-based approach facilitating community decision-making on information sharing. Using cellular automata as a computational model, the proposed game design enables dynamic simulations of evolving consent scenarios within a community context. Players, representing community members, navigate through decision spaces influenced by changing factors, contributing to a collective understanding of preferences and concerns related to information sharing. Key objectives include designing game mechanics that simulate cellular automata processes for an intuitive representation of evolving community consent scenarios, implementing dynamic decision spaces that adapt to changing factors, fostering community engagement through the game design, addressing privacy and security considerations, and incorporating user feedback mechanisms for iterative improvements. The research outcomes aim to contribute to a novel paradigm for seeking community consent by combining game design and cellular automata, providing a dynamic and interactive platform for community members to actively engage in the decision-making process surrounding information sharing. The insights gained from the game-based simulations have broader implications for designing participatory consent mechanisms in various contexts, aligning with evolving standards of ethical and transparent information governance.

Minimum No. of Hires:	12
Expected No. of Hires:	12
Location(s)/Place of Posting/Online:	Hyderabad
Skillset:	Research
Students with backlog eligible:	No

Selection Process

Resume Shortlist:	Yes
Mode of Selection:	Hybrid
Resume shortlisting before test?:	No
Test:	No
Group Discussion:	No
Personal Interview:	Yes
Technical Round:	No
HR Round:	No
Other Processes (if any):	
No. of Rounds:	2
Medical Test:	No

Eligible Academic Programs

Diversity Recruiting:	No
Eligible Years:	Graduating in 2027 (Third Year Students) - Dual Degree, Graduating in 2026 (Pre-Final Year Students) - B.Tech / Dual / Master's , Graduating in 2027 (Second Year Students) - B.Tech / Dual
Eligible Departments:	B.Tech in Biochemical Engineering & Biotechnology, B.Tech in Chemical Engineering, B.Tech in Civil Engineering, B.Tech in Computer Science & Engineering, B.Tech in Electrical Engineering, B.Tech in Electrical Engineering (Power and Automation), B.Tech in Energy Engineering, B.Tech in Engineering Physics, B.Tech in Engineering and Computational Mechanics, B.Tech in Materials Engineering, B.Tech in Mathematics & Computing, B.Tech in Mechanical Engineering, B.Tech in Production & Industrial Engineering, B.Tech in Textile Engineering, B.Tech and M.Tech in Biochemical Engg & Biotechnology, B.Tech and M.Tech in Chemical Engineering, B.Tech and M.Tech in Computer Science & Engineering, B.Tech and M.Tech in Mathematics & Computing, B.Tech and M.S.(R) in Electrical Engineering

Stipend Details

Stipend (per month) (In INR Per Month):	30,000 INR Per Month
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